Nikolaos Giakoumoglou

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EDUCATION

Imperial College London — London, United Kingdom
PhD in Electrical and Electronic Engineering — Jan 2024 – Present

- Supervisor: Prof. Tania Stathaki, Communications and Signal Processing Group
- Full PhD scholarship, funded by the Department of Electrical and Electronic Engineering

Aristotle University of Thessaloniki — Thessaloniki, Greece
Integrated MSc / Diploma in Electrical and Computer Engineering — Sep 2016 – Nov 2021

- GPA: 8.91/10; graduated in the top 5% of the class
- Thesis: "Plaque Classification in Carotid Ultrasounds", supervised by Prof. Anastasios Ntelopoulos

RESEARCH EXPERIENCE

Imperial College London, Communications and Signal Processing Group — London, United Kingdom
Research Postgraduate / PhD Researcher — Jan 2024 – Present

- Research on self-supervised learning, knowledge distillation, joint-embedding predictive architectures

Centre for Research and Technology Hellas, Information Technologies Institute — Thessaloniki, Greece
Research Assistant — Feb 2022 – Dec 2023

- Developed and deployed deep learning computer vision systems for real-world agricultural robotics
- Led and managed AI research and development activities within EU Horizon 2020 projects

RECENT PUBLICATIONS

1. **N. Giakoumoglou**, *et al.*, "An Empirical Study of Training Self-Supervised Vision Transformers with Synthetic Hard Negatives," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2027.
2. **N. Giakoumoglou**, *et al.*, "A Stochastic Latent Variable Generalization of Joint-Embedding Predictive Architectures," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2027.
3. **N. Giakoumoglou**, *et al.*, "Generalizable AI-Generated Image Detection with Embedding-Space Explanations," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2027.
4. **N. Giakoumoglou**, *et al.*, "What Makes Synthetic Hard Negatives Work in Vision-Language Pretraining?," in *Asian Conference on Computer Vision (ACCV)*, 2026.
5. K. M. Papadopoulos, **N. Giakoumoglou**, *et al.*, "Expert Clustering and Knowledge Transfer for Whole Slide Image Classification," in *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2026. **Oral**.
6. L. Iijima, **N. Giakoumoglou**, and T. Stathaki, "A Multimodal Approach for Cross-Domain Image Retrieval," in *International Conference on Computer Vision Theory and Applications (VISAPP)*, 2026. **Oral**.
7. **N. Giakoumoglou** and T. Stathaki, "Cluster Contrast for Unsupervised Visual Representation Learning," in *IEEE International Conference on Image Processing (ICIP)*, 2025.

More publications can be found at Google Scholar and my Personal Webpage.

TEACHING AND VOLUNTEERING

STEP-UP Digital Research Champion, Imperial College London — Jul 2025 – Present
Student Representative, Communications and Signal Processing Group — Sep 2024 – Present
Graduate Teaching Assistant, Imperial College London — Sep 2024 – Present

AWARDS

EEE Departmental PhD Scholarship: Full PhD scholarship and funding — Jan 2024

INVITED TALKS

Google DeepMind — Online, Oct 2026
Invited research talk on self-supervised visual representation learning and Joint-Embedding Predictive Architectures

Imperial Research Computing Showcase Day 2025 — London, United Kingdom, 2025
Invited talk on large-scale medical imaging, model pretraining, and high-performance computing (HPC)

MEMBERSHIPS

IEEE Signal Processing Society (SPS); Student Member — Jun 2025 – Present
Institute of Electrical and Electronics Engineers (IEEE); Student Member — Mar 2023 – Present
Technical Chamber of Greece; Member — Feb 2022 – Present

MISCELLANEOUS

Service: Reviewer for NeurIPS, ICML, ICLR, CVPR, ECCV, AAAI, WACV, ACCV, MLSP
Technical skills: Python, PyTorch, Linux, Git, High-Performance Computing (HPC)